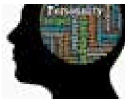
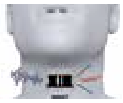


New personality analysis test
Rsearchers are now using data from phone accelerometers to predict people's personalities. **https://dgit.in/aug19-11**



Artifical throat
A wearable artificial throat is being developed which can transform throat movements into actual sounds. **https://dgit.in/aug19-12**

OPTOGENETICS

An emerging science of controlling cells using a combination of light and genetics can pave the way for novel treatments for various disorders.

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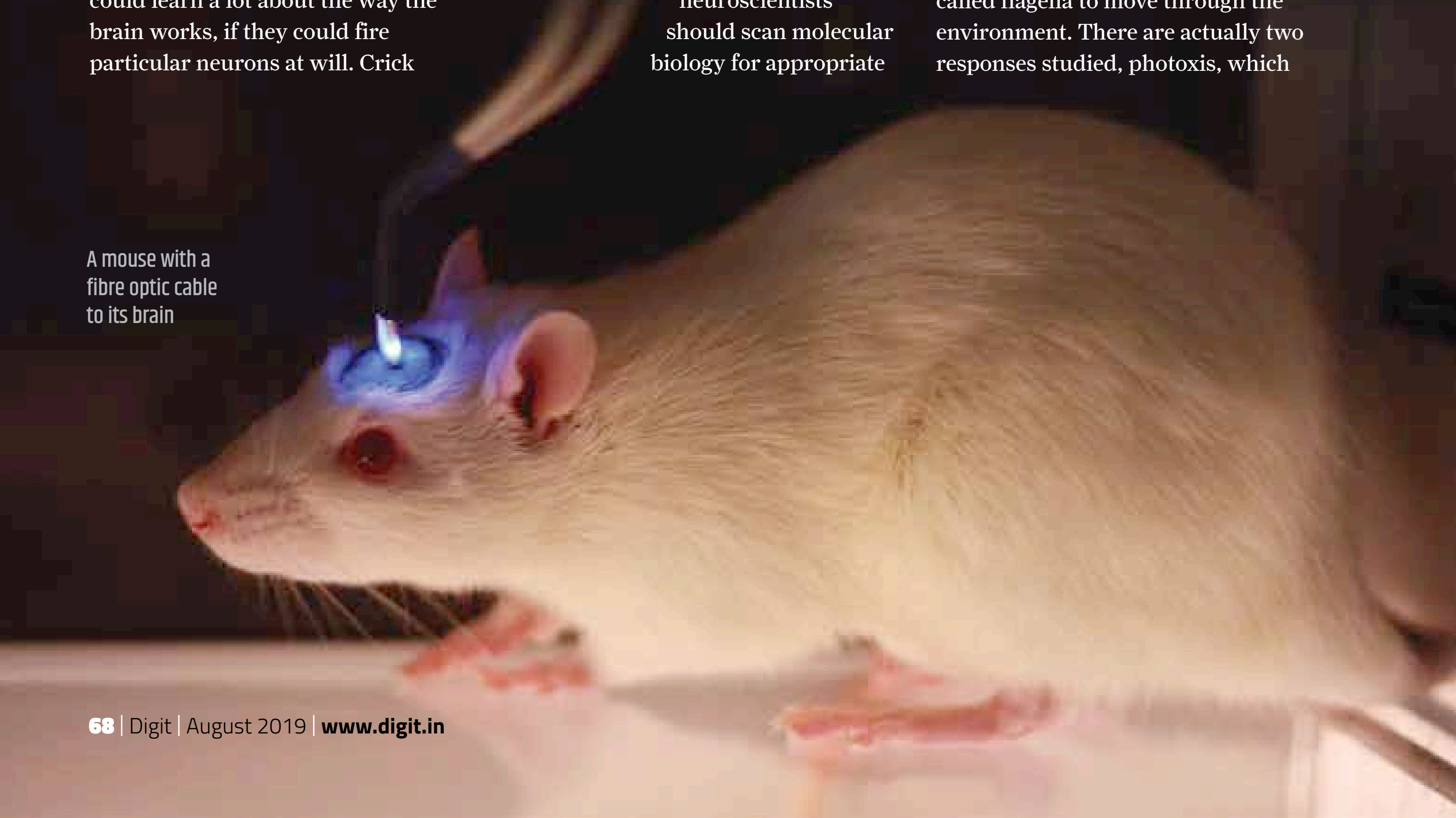
In 1979, Francis Crick, the co-discoverer of the double helix structure of DNA, wrote an article in Scientific American outlining the need for a faster way to directly control particular neurons within the brains of living organisms. The available methods at that time were electrical stimulation and drugs. Both these approaches were slow to work, and were spatially diffused, that is all the cells in the vicinity of the electrode or drug responded. There was a lack of precision, and researchers could learn a lot about the way the brain works, if they could fire particular neurons at will. Crick speculated that it could one day be possible for molecular biologists to control neurons using light, which can easily be turned on and off rapidly. If the neurons were made sensitive to light, then several advancements could be made in the field of neuroscience. In 1999, 20 years later, the problem still persisted. Crick wrote another paper exploring possible ways to use developments in genetics, to advance the field of neuroscience. He concludes the paper by noting, "The point I want to stress is that neuroscientists should scan molecular biology for appropriate techniques but, most important, they should ask their molecular biology friends for new tools. They should tell them what their difficulties are and what they want to do. Once the word gets around that a certain type of problem exists it is surprising how often someone has a bright idea of how to solve it. So, don't be shy—ask!"

LIGHT IN THE BRAIN

Just a few years later, biologists had that bright idea. It had been known for decades that some algae swim towards or away from light. They would use whip like appendages called flagella to move through the environment. There are actually two responses studied, photoxis, which



A mouse with a fibre optic cable to its brain